Mathematics practice booklet: Grade 10 expectations

Grade 10 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 10 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-APR.A.1, A-SSE.B.3, A-REI.B.4, F-IF.C.7, F-IF.C.8

A worked example

Solve by factoring: x^2 - 2x - 15 = 0. Then check one answer.

1. Look for two numbers that multiply to -15 and add to -2: they are -5 and 3.
2. Factor: (x - 5)(x + 3) = 0.
3. A product is zero only when one factor is zero, so x - 5 = 0 or x + 3 = 0.
4. Check x = 5: 25 - 10 - 15 = 0. It works.

x = 5 or x = -3

Questions

1. Solve by factoring: x^2 - 5x + 6 = 0

2. Solve by factoring: x^2 + 3x - 10 = 0

3. Solve: 2x^2 - 8 = 0

4. Solve by completing the square: x^2 + 6x + 2 = 0

5. Use the quadratic formula to solve x^2 - 4x - 1 = 0.

6. Use the quadratic formula to solve 2x^2 + 3x - 2 = 0.

7. Find the discriminant of x^2 + 2x + 5 = 0. How many real solutions does it have?

8. Expand and simplify: (x + 3)(2x - 5)

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9. Let A = 3x^2 + 2x - 1 and B = x^2 - 5x + 4. Find A + B and A - B.

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10. For y = x^2 - 6x + 5, find the x-intercepts, the vertex and the y-intercept.

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11. The rectangular floor of a majlis is 3 m longer than it is wide, and its area is 40 square m. Find its width and length.

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12. Omar kicks a ball upward. Its height is h = -5t^2 + 20t metres after t seconds. When does it land, and what is its greatest height?

Section 2: Shapes, measuring and space

Standards practised: G-CO.B.8, G-CO.C.9, G-CO.C.10, G-SRT.B.5, G-SRT.C.6, G-SRT.C.8, G-C.A.2, G-C.B.5

A worked example

A ramp is 10 m long and rises at an angle of 20 degrees to the ground. How high does it rise? (sin 20 degrees is about 0.342)

1. The ramp is the hypotenuse. The height is the side opposite the 20 degree angle.
2. Opposite and hypotenuse go with sine: sin 20 = height / 10.
3. Height = 10 x 0.342 = 3.42.

About 3.4 m

Questions

1. Name the criteria that prove two triangles congruent. Is two sides and a non-included angle (SSA) enough?

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2. In triangles ABC and DEF, AB = DE = 6 cm, angle B = angle E = 50 degrees and BC = EF = 9 cm. Which criterion shows they are congruent?

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3. Two straight lines cross. One of the four angles is 125 degrees. Find the other three angles and say which are equal because they are vertical angles.

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4. An isosceles triangle has a top (apex) angle of 40 degrees. Find the base angles and explain why they are equal.

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5. Triangle ABC has sides 3, 4 and 5 cm. Triangle DEF is similar with sides 9, 12 and x cm. Find x and the scale factor.

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6. A lamp post 3 m tall casts a shadow 2 m long. At the same time a palm tree casts a shadow 10 m long. How tall is the palm tree, and why can you use similar triangles?

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7. In a right triangle the side opposite angle A is 6 cm and the hypotenuse is 10 cm. Find sin A and the size of angle A to the nearest degree.

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8. Khalid stands 50 m from the base of a tower and sees the top at an angle of elevation of 35 degrees. Find the height of the tower. (tan 35 degrees is about 0.700)

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9. A rectangular courtyard is 12 m long and 5 m wide. How long is a walk along its diagonal?

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10. A circular garden has radius 7 m. Find its circumference and area in terms of pi, then to one decimal place. A quarter of the garden is a sector: find its arc length and its area to one decimal place.

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11. In a circle, a central angle of 80 degrees and an inscribed angle both stand on the same arc. Find the inscribed angle. What is an inscribed angle that stands on a diameter?

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12. A point P is 13 cm from the centre of a circle of radius 5 cm. A tangent from P touches the circle at T. Find the length PT, and explain how you know the angle at T.

Section 3: Data, graphs and chance

Standards practised: S-CP.A.1, S-CP.A.2, S-CP.A.3, S-CP.A.4, S-CP.B.6, S-CP.B.9

A worked example

A jar holds 3 red and 2 green sweets. Hind takes two sweets without putting the first one back. Find the probability that both are red.

1. The first sweet is red with probability 3/5.
2. Now 2 red sweets remain among 4 sweets, so the second is red with probability 2/4.
3. Multiply: 3/5 x 2/4 = 6/20 = 3/10.

3/10

Questions

1. A coin is tossed and a die is rolled. Find the probability of heads and a 4.

2. P(A) = 0.4, P(B) = 0.5 and P(A and B) = 0.2. Are A and B independent? Explain.

3. A bag has 5 red and 3 blue counters. Two counters are drawn, and the first is put back before the second. Find the probability that both are red.

4. Use the same bag, but this time the first counter is NOT put back. Find the probability that both are red.

5. 50 students were asked whether they prefer swimming or running. Boys: 15 swimming, 10 running. Girls: 20 swimming, 5 running. How many prefer swimming, and what is P(swimming) for a student chosen at random?

6. In the same survey, a girl is chosen at random. Find the probability that she prefers swimming.

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7. In the same survey, a student who prefers swimming is chosen at random. Find the probability that it is a girl.

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8. Using the same survey, are being a girl and preferring swimming independent? Explain.

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9. P(A) = 0.5 and P(A and B) = 0.3. Find P(B given A).

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10. Six runners are in a school race. In how many different ways can gold, silver and bronze be given?

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11. How many different ways can the 4 letters of OMAN be arranged in a row? If they are arranged at random, what is the probability of spelling OMAN?

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12. The chance that a metro train is on time on any given day is 0.9, and days are independent. Find the probability that it is on time on both Sunday and Monday, and the probability that it is late at least once.

Section 4: Numbers and calculating

Standards practised: N-RN.A.1, N-RN.A.2, N-RN.B.3, N-Q.A.1, N-Q.A.3

A worked example

Simplify the square root of 48 plus the square root of 12.

1. Write each as a perfect square times another number: √48 = √16 x √3 = 4√3.
2. √12 = √4 x √3 = 2√3.
3. These are like terms, so add them: 4√3 + 2√3 = 6√3.

6√3

Questions

1. Simplify: √72

2. Simplify: √50 + √18

3. Simplify: √3 x √12

4. Write 6 / √3 with a whole-number denominator.

5. Evaluate: 49^(1/2) and 8^(1/3)

6. Evaluate: 27^(2/3) and 4^(-3/2)

7. Write the cube root of 5 using a rational exponent, and write 7^(2/5) using a radical.

8. Is 1/2 + √3 rational or irrational? Is √2 x √8 rational or irrational?

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9. A car travels at 108 km per hour. Convert this speed to metres per second.

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10. A desalination plant makes 3 x 10^4 cubic m of water a day, and 1 cubic m is 10^3 litres. How many litres is that in 250 days? Give the answer in scientific notation.

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11. A tank really holds 250 litres. Hamdan measures it as 245 litres. Find the percent error.

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12. A solar park is 2 km long and 500 m wide. Find its area in square metres and in square kilometres. Why should the units match before you multiply?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. x = 2 or x = 3 ((x - 2)(x - 3) = 0)
2. x = -5 or x = 2 ((x + 5)(x - 2) = 0)
3. x = 2 or x = -2 (x^2 = 4)
4. x = -3 + √7 or x = -3 - √7 ((x + 3)^2 = 7)
5. x = 2 + √5 or x = 2 - √5 (discriminant 16 + 4 = 20, and √20 = 2√5)
6. x = 1/2 or x = -2 (discriminant 9 + 16 = 25, so x = (-3 ± 5) / 4)
7. Discriminant = 4 - 20 = -16, which is negative, so there are no real solutions.
8. 2x^2 + x - 15 (2x^2 - 5x + 6x - 15)
9. A + B = 4x^2 - 3x + 3. A - B = 2x^2 + 7x - 5 (subtract every term of B).
10. x-intercepts: x = 1 and x = 5. Vertex: (3, -4). y-intercept: (0, 5).
11. Width 5 m, length 8 m (w(w + 3) = 40 gives (w + 8)(w - 5) = 0, and width cannot be negative).
12. It lands at t = 4 s (-5t(t - 4) = 0). The greatest height is 20 m, at t = 2 s, the vertex of the graph.

Section 2: Shapes, measuring and space

1. SSS, SAS, ASA, AAS, and HL for right triangles. SSA is not enough, because it can fit two different triangles.
2. SAS: two sides and the angle between them are equal.
3. The vertical angle is also 125 degrees. The other two are 55 degrees each (180 - 125).
4. 70 degrees each ((180 - 40) / 2). In an isosceles triangle the angles opposite the equal sides are equal.
5. x = 15; scale factor 3 (9 / 3 = 3 and 5 x 3 = 15).
6. 15 m (3 / 2 = h / 10). The sun makes the same angle with the ground for both, so the two triangles have equal angles (AA).
7. sin A = 6 / 10 = 0.6, so A is about 37 degrees.
8. About 35 m (50 x 0.700 = 35)
9. 13 m (144 + 25 = 169)
10. Circumference 14 pi, about 44.0 m. Area 49 pi, about 153.9 square m. Quarter sector: arc length 7 pi / 2, about 11.0 m; area 49 pi / 4, about 38.5 square m.
11. 40 degrees (half of the central angle). An angle inscribed in a semicircle is 90 degrees.
12. 12 cm. A tangent is perpendicular to the radius at T, so the triangle is right-angled and 13^2 - 5^2 = 144.

Section 3: Data, graphs and chance

1. 1/12 (1/2 x 1/6, the events are independent)
2. Yes: 0.4 x 0.5 = 0.2, which equals P(A and B).
3. 25/64 ((5/8) x (5/8))
4. 5/14 (5/8 x 4/7 = 20/56)
5. 35 students; P(swimming) = 35/50 = 7/10.
6. 20/25 = 4/5 (P(swimming given girl))

7. 20/35 = 4/7 (P(girl given swimming))
8. No: P(swimming) = 0.7 but P(swimming given girl) = 0.8. They are different, so the events are not independent.
9. 0.6 (0.3 / 0.5)
10. 120 ways (6 x 5 x 4)
11. 24 ways (4 x 3 x 2 x 1). The probability is 1/24.
12. Both on time: 0.9 x 0.9 = 0.81. Late at least once: 1 - 0.81 = 0.19.

Section 4: Numbers and calculating

1. 6√2 (√36 x √2)
2. 8√2 (5√2 + 3√2)
3. 6 (√36)
4. 2√3 (multiply top and bottom by √3 to get 6√3 / 3)
5. 7 and 2
6. 9 and 1/8 (27^(2/3) = 3^2 = 9; 4^(-3/2) = 1 / 2^3)
7. 5^(1/3). 7^(2/5) is the fifth root of 7^2 (the fifth root of 49).
8. 1/2 + √3 is irrational (rational plus irrational). √2 x √8 = √16 = 4, which is rational.
9. 30 m/s (108 x 1000 / 3600)
10. 7.5 x 10^9 litres (3 x 10^7 per day, times 2.5 x 10^2 days)
11. 2% (error 5 litres, and 5 / 250 = 0.02)
12. 1,000,000 square m, which is 1 square km (2000 x 500). If one length is in km and the other in m, the product would mix units and give a wrong answer.